

Project Progress - Week 1 of March 2025

Joan Ronquillo

March 4, 2025

This is a file that contains the tracking of the activities related to the OFC project during the first week of March 2025. The activities are divided into groups and a summary of the progress is provided.

1 Initial Status of the Project

The project has made significant progress up until the first week of March 2025. The key achievements include:

- Successful implementation and validation of the optical force calculation function in the dipole approximation.
- Development of test and validation scripts that confirm the correct dependence on the permittivity.
- Advancement in the theoretical documentation, particularly in the section on polarizability.
- Detailed derivation of the quasi-static polarizability formula for a dielectric sphere.

2 Potential Tasks for the Week

The following tasks have been identified as potential areas of focus for the first week of March 2025:

- Optimize the functions for querying the refractiveindex.info database.
- Complete the integration of permittivities into an specific equation.
- Finish documenting and providing examples of usage.

3 Week Tracking

3.1 Monday - March 3, 2025

The repository structure has been fixed and the management of Python modules has been addressed. Additionally, I have been learning how to compile LaTeX projects located inside the repository. This document serves as an example of the compilation process.

3.2 Tuesday - March 4, 2025

The tasks for the week have been specified and the progress in documentation, code, and testing has been tracked.